Table 10.1: Soil Microorganisms with example

Soil Microorganisms	Examples
Bacteria	*Agrobacterium* *Bacillus* *Clostridium* *Pseudomonas*
Actinomycetes	*Actinomyces* *Nocardia* *Streptomyces*
Fungi	*Aspergillus* *Fusarium* *Alternaria* *Cladosporum*
Soil algae	*Anabaena* *Oscillatoria* *Nostoc*
Protozoa	*Colpoda* *Nematodes* *Pleurotricha* *Heteromita*
Bacteriophages	T4 Bacteriophages



One teaspoon of productive soil can contain between 100 million and 1 billion bacteria. These living microorganisms recycle organic material, promoting soil fertility and supporting plant growth. By practicing conservation tillage, farmers can maintain biodiversity in their soil.

Soil Bacteria

Among the soil microorganisms, bacteria are most dominant group of organisms. All kinds of bacteria are found in soil. This is because all kinds of organic refuse are disposed on the soil

Many of the soil bacteria perform useful functions like decomposition of organic matter, conversion of soil constituents into useful materials, production of antibiotics in the soil and biogeochemical cycling of elements like Carbon, Nitrogen, Phosphorus, Iron, Sulphur and Manganese. The bacterial population of the soil exceed the population of all other groups of microorganisms in both number and variety.

Soil Actinomycetes

The actinomycetes population is present as many as millions per gram of soil. The most predominant genera present in the soil are *Nocardia, Streptomyces* and *Micromonospora*. Actinomycetes are capable of degrading many complex organic substances and therefore play an important role in building soil fertility. One of the most notable characteristics of the actinomycetes is their ability to produce antibiotics. Examples: Streptomycin, neomycin, erythromycin and tetracycline.

Soil Fungi

Next to bacterial population in soil, fungi dominates in all kinds of soil. It possess filamentous mycelium composed of individual hyphae. All environmental factors which influence the distribution of bacteria and actinomycetes also influence the fungal flora of soil. The quality and quantity of organic matter present in the soil have a direct influence on the fungal numbers in soil. Fungi are dominant in acidic soils because acidic environment is not supportive for the existence of either bacteria or actinomycetes.

TN_GOVT_XI_Micro_Biology_CH10.indd 152 16-02-2019 10.03.51 AM